

Texas Mineral Rights: Harvest Proof & Data Limits

Pilot research for location-intelligence (July 2026). Goal: confirm whether mineral ownership and related O&G records can be harvested from primary sources, document limits (rate, legal, pay, hard), and collect working samples where feasible without large-scale scraping.

Executive Summary

1. RRC (Railroad Commission) data is broadly obtainable for free via bulk MFT downloads and ArcGIS REST - wells, permits, production, pipelines, UIC, etc. It does NOT include private mineral ownership or courthouse title chains.
2. Courthouse mineral ownership lives in county clerk real-property indexes (deeds, reservations, O&G leases, probates). Index search is public; bulk index+images are sold county-by-county.
3. GovOS/Kofile portals (publicsearch.us) require client-side JavaScript - no public REST API. DEMONSTRATED: Playwright browser harvest pulled 150 index rows from Reeves + Midland (16 mineral-related).
4. No fundamental hard limit prevents obtaining Texas public records, but 254 counties x disparate vendors means statewide ownership is an aggregation problem, not a single-API problem.

Samples Collected (This Pilot)

File	Size	Description	Status
courthouse_mineral_index_harvest.csv	150 rows	Reeves + Midland GovOS index via Playwright	OK
courthouse_mineral_only_sample.csv	16 rows	Mineral deed/lease/conveyance subset	OK
rrc_midland_wells_sample.csv	100 rows	RRC MFT api165.dbf (Midland wells)	OK
rrc_gis_midland_wells_sample.csv	200 rows	RRC ArcGIS Well Locations - Midland bbox	OK
courthouse_reeves_mineral_doctypes.csv	644 types	Mineral/O&G doc-type codes from Reeves portal	OK

Re-run courthouse harvest: `python scripts/harvest_courthouse_demo.py` (requires playwright install chromium).

Example row: NEECE JOHN R -> BRIGHAM MINERALS LLC, MINERAL DEED, doc# 2019017441, Reeves County.

RRC MFT: use PrimeFaces ajax POST (`scripts/mft_ajax_download.py`). Contact: digital@rrc.texas.gov.

What the RRC Provides (Full Catalog)

Free bulk datasets at rrc.texas.gov/resource-center/research/data-sets-available-for-download/

Category	Contents	Update
Digital Map Data	Well/Survey/Pipeline/Base shapefiles by county; Statewide API ASCII/DBF	2x/week
Drilling Permits	Master, trailer, daily/monthly pending; W1 images; horizontal permits	Daily-Monthly
Oil & Gas Field Data	Field names, rules, annual reports, statewide field EBCDIC	Monthly
Production Data	Oil/gas ledgers by district; historical ledgers; CSV query dumps	Monthly
Regulatory	P-4 certs, dockets, P5 operators, R3 gas plants (JSON), Oil/ICE weekly	Weekly-Monthly
Well Data	Full Wellbore (ASCII/EBCDIC ~366MB gz); wellbore query; 26-mo status; completions	Weekly-Monthly
Tax Incentive	High-cost gas, NGPA, ST-1 CSV	Monthly
UIC	Injection well inventory ASCII/EBCDIC	Monthly
GIS REST	gis.rrc.texas.gov - wells, pipelines, surveys, districts (max 1000/query)	Live

What RRC Does NOT Provide

- Private mineral ownership, royalty fractions, or deed chains
- Surface/mineral severance from county records
- Coal (Texas RRC is oil/gas focused; coal is rare in TX; check GLO/federal for lignite)
- Legal title opinions - RRC disclaimers state datasets are informational, not authoritative records

RRC Limits Matrix

Channel	Rate/Technical	Cost	Legal/Policy	Hard limit?
Bulk MFT downloads	None published	Free	Public records - intended path for volume	No
Online query system (webapps)	Sessions terminated if automated	Free	Site policy bans robots causing degradation	Soft - use bulk instead
ArcGIS REST	maxRecordCount 1000/query	Free	Fair use; throttle politely	No
EBCDIC/mainframe files	Format conversion burden	Free	User must convert	No
Coal/mineral title	Not in RRC scope	n/a	Wrong agency for ownership	Hard for ownership

Courthouse Mineral Ownership - Where It Lives

Recorded in County Clerk Official Public Records: mineral deeds, reservations, O&G leases, assignments, royalty deeds, probates, affidavits of heirship, plats. Each of 254 counties maintains its own index; vendors include GovOS (publicsearch.us), Tyler EagleWeb, Avenu (TexasLandRecords), Harris custom portal, Kofile legacy systems.

Courthouse Harvest Proof (Demonstrated)

CONFIRMED via Playwright browser automation on GovOS portals (July 2026):

- 150 index rows harvested from Reeves + Midland (50 per search x 3 searches)
 - 16 mineral-related instruments (MINERAL DEED, MINERAL CONVEYANCE, O&G LEASE, etc.)
 - Fields: grantor, grantee, doc_type, recorded_date, doc_number, legal_description
 - Method: navigate to /results?searchQuery=..., wait for JS render, parse results table
 - Plain HTTP requests alone fail; browser automation works for small-scale pilot harvest
- Reeves also preloads docTypeMappings (644 mineral/O&G codes). Clerk: 432-287-0222.

Courthouse Limits Matrix

Channel	Rate/Technical	Cost	Legal/Policy	Hard limit?
GovOS index search (manual)	Unpublished; no API key	Free index	Public records; vendor ToS on images	No for index
Browser automation (Playwright)	Demo: 150 rows/county search	Free index	Pilot proven on GovOS; scale per-county	Soft
Document images	Per-page fees	\$1/pg typical	GovOS: copies priced; Harris watermarked free online	No
Bulk index (pipe-delimited)	County policy	Varies - see below	PIA (Gov Code 552) + Local Gov 195.007	No
Bulk images / FTP	County policy	Reeves \$50/wk, \$200/mo FTP	Direct from clerk - preferred source	No
Statewide ownership API	Does not exist	Vendor subscriptions	CourthouseDirect, Enverus, TexasFile	Hard - no single government API
Pre-1980s unindexed records	In-office/microfilm	Staff search fees	Not all backfiles digitized	Partial hard gap

Bulk / Offline County Data (Published - Not Yet Contacted)

County	Phone	Bulk Option	Portal	Notes
Reeves	432-287-0222	FTP weekly \$50 / monthly \$200	reeves.tx.publicsearch.us	Fee schedule on county site
Harris	713-274-6390	Custom date-range index (pipe txt) + TIFF; monthly FTP	cclerk.hctx.net/PublicRecords.aspx	Data Sales desk
Fort Bend	County Clerk	FTP daily \$10/GB + \$50 setup	fortbendcountytx.gov	Computer Information Fee Schedule
Midland	432-688-4401	GovOS portal; bulk terms not published online	midland.tx.publicsearch.us	Call clerk for PIA/bulk

Draft Clerk Outreach (Ready to Send)

Subject: Public Information Act / Bulk OPR Index Request - Mineral & O&G Instruments

Dear County Clerk Records Division,

We are researching official public records for mineral and oil & gas title work statewide. Could you advise:

1. Do you offer bulk export of the Official Public Records index (grantor, grantee, doc type, recorded date, instrument number, legal description) in CSV or pipe-delimited text?
2. Is FTP or SFTP available for recurring index and/or TIFF/PDF images? Fee schedule?
3. What date range is fully indexed online vs. in-office only?
4. For PIA requests, what is your per-page/per-file cost cap and typical turnaround?

We prefer going directly to the county rather than third-party aggregators.

Thank you,

[Your name / organization]

Hard vs Soft Limits - Can a Million Computers Get It All?

Soft limits (scale with money, time, or polite engineering):

- RRC online query throttling - bypass via free bulk MFT
- GovOS bot detection - bypass via bulk FTP purchase or PIA
- 254 county schemas - bypass via vendor aggregation (\$\$) or phased county rollout
- Image costs at \$1/page - bypass via bulk image subscriptions

Hard / structural limits:

- No statewide mineral ownership register exists in Texas law
- Ownership must be reconstructed from chains of title, not queried like well API numbers
- Unrecorded leases/title gaps are inherently invisible in any public index
- Federal minerals (BLM) and state minerals (GLO) are separate from county OPR
- Pre-digitization records may exist only in books/microfilm with manual lookup

Conclusion: Volume is solvable; complete automated ownership for all tracts is not - title research remains interpretive even with full index access.

Recommended Next Steps

1. Wire harvest_courthouse_demo.py into pipeline for pilot counties (Reeves, Midland, Tarrant)
2. Purchase Reeves weekly FTP (\$50) for bulk index without per-search browser overhead
3. Send PIA/bulk inquiry to Midland + Harris Data Sales (713-274-6390)
4. Integrate RRC well data with courthouse leases by legal description (expect fuzzy joins)
5. Scale to additional GovOS counties using same Playwright pattern

Sample File Locations

Project: C:\Users\kkr84\Projects\location-intelligence

data/samples/courthouse_mineral_index_harvest.csv - 150 courthouse index rows

data/samples/courthouse_mineral_only_sample.csv - 16 mineral-related rows

data/samples/rrc_midland_wells_sample.csv - 100 RRC wells from MFT DBF

scripts/harvest_courthouse_demo.py - re-run courthouse harvest demo

reports/Texas_Mineral_Rights_Data_Strategy.pdf - three-tier build/buy plan